

A. Let us consider the case of Good A in Vietnam. The demand and supply functions for Good A in Vietnam are,

$$Q_d = 250 - 2P \text{ and } Q_s = 10 + P.$$

a) If the world price is 60, then is Vietnam an importer or exporter?

Answer a. [Find out equilibrium price, if it is more than world price they will import and if it is less than world price then they will export]

At equilibrium, $Q_D = Q_S = Q^*$

$$250 - 2P^* = 10 + P^*$$

$$3P^* = 240$$

$$P^* = 80$$

Here, equilibrium price (80) is greater than the world price (60), so Vietnam will be an importer.

b) Find out the quantity of imports at $P_w = 60$?

Answer b. [Find Imports = $(Q_d - Q_s)$ at $P_w = 60$]

The quantity of imports [at $P_w = 60$] = $Q_D - Q_S$ [at $P_w = 60$]

The quantity of imports [at $P_w = 60$] = $\{250 - 2(60)\} - \{10 + 60\}$

The quantity of imports [at $P_w = 60$] = $130 - 70$

The quantity of imports [at $P_w = 60$] = 60

c) If the import quota was 45 units, what will be the new price in the Vietnamese market?

Answer c. [Find price at $Q_d - Q_s = 45$]

$$Q_D - Q_S = 45$$

$$\{250 - 2P\} - \{10 + P\} = 45$$

$$250 - 2P - 10 - P = 45$$

$$240 - 3P = 45$$

$$3P = 195$$

$$P = 65$$

If the import quota was 45 units, the new price in the Vietnamese market will be 65.

d) How much more/less of Good A will be produced domestically?

Answer d. [**Qs after quota - Qs before quota**]

$$Qs \text{ before quota } (P_w=60) = 10 + P = 10 + 60 = 70$$

$$Qs \text{ after quota } (P=65) = 10 + P = 10 + 65 = 75$$

$$\text{Therefore, } 75 - 70 = 5$$

5 more units of Good A will be produced domestically.

e) How much more/less of Good A will be consumed domestically?

Answer e. [**Qd after quota - Qd before quota**]

$$Qd \text{ before quota } (P_w=60) = 250 - 2P = 250 - 2(60) = 130$$

$$Qd \text{ after quota } (P=65) = 250 - 2P = 250 - 2(65) = 120$$

$$\text{Therefore, } 120 - 130 = 10$$

10 less units of Good A will be consumed domestically.

f) What is the percentage change in imports after the imposition of import quota?

Answer f. (Use conventional method) -

The percentage change in imports after the imposition of import quota =

$$[\{(\text{Import after quota} - \text{Import before quota}) / \text{Import before quota}\} \times 100]$$

Import before quota at world price = 60

Import after quota = 45

Therefore, $\{(45-60)/60\} \times 100 = -25\%$

The percentage change in imports after the imposition of import quota is 25%.

g) If the quantity demanded before quota was 355 units what will be the new quantity demanded if it decreases by 20% or 1/5th?

Answer g. Percentage change in imports after the imposition of import quota = 20%

Amount of decrease in quantity demanded after quota = $355 \times (20/100) = 71$

Import after quota = Import before quota - Amount decreased after quota = $355 - 71 = 284$

Therefore, the new quantity demanded after quota is 284.

h) If the quantity demanded before quota was 355 units what will be the new quantity demanded if it decreases by 14%?

Answer h. Percentage change in imports after the imposition of import quota = 14%

Amount of decrease in quantity demanded after quota = $355 \times (14/100) = 49.7$

Import after quota = Import before quota - Amount decreased after quota = $355 - 49.7 = 305.3$

Therefore, the new quantity demanded after quota is 305.3.

1. The existence of any "gains from trade" relies on
A) closed economies.
B) absolute advantage.
C) **comparative advantage.**
D) both absolute and comparative advantage.
E) tariffs.

2. Comparative advantage implies that a country will
A) **export those goods in which the country has a comparative advantage.**
B) find it difficult to conclude free trade agreements with other nations.
C) import those goods in which the country has a comparative advantage.
D) export goods produced by domestic industries with low wages relative to its trading partners.

3. There will be no gains from specialization and trade between two countries if
A) neither country has an absolute advantage in the production of any good;
B) neither country has a comparative advantage in the production of any good;
C) opportunity costs are the same in the two countries.
D) Both A and B
E) **Both B and C**

4. After the United States imports a good, production in the United States _____ and consumption in the United States _____.
A) **decreases; increases**
B) increases; decreases
C) increases; increases
D) decreases; decreases

5. A tariff
a) Increases the volume of trade.
b) **Reduces the volume of trade.**
c) Has no effect on the volume of trade.
d) Both (a) and (c).

6. Who benefits from a tariff on a good?
A) **Domestic producers of the good**
B) Foreign producers of the good
C) Domestic consumers of the good
D) Foreign governments

7. A tariff on imported peanuts _____ the quantity of peanuts imported and _____ the domestic price of peanuts.
A) increases; lowers
B) decreases; decreases
C) **decreases; increases**
D) does not change; increases

8. A country opens up to trade and becomes an importer of sugar. In the sugar market, consumer surplus will _____, producer surplus will _____, and total surplus will _____.

- A) increase; decrease; decrease
- B) **increase; decrease; increase**
- C) decrease ; increase; increase
- D) decrease; decrease; decrease

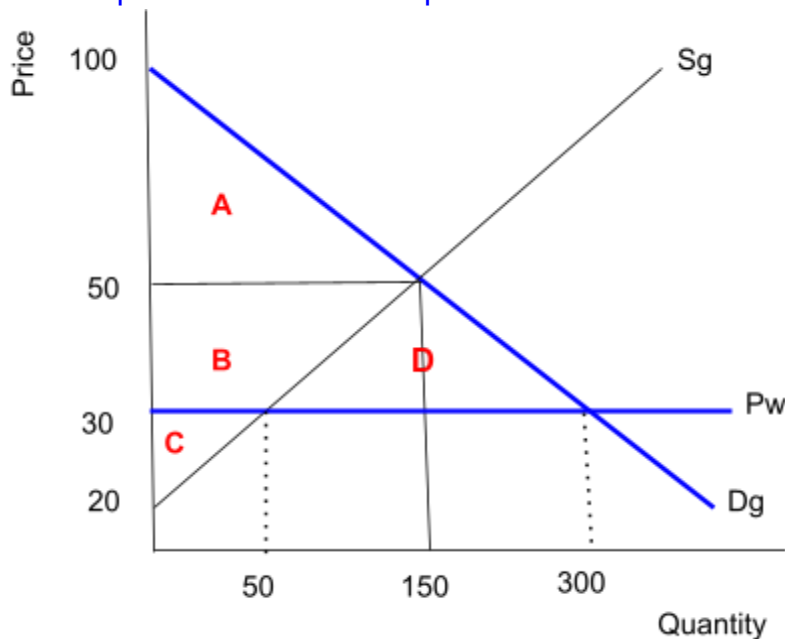
9. A country opens up to trade and becomes an exporter of sugar. In the sugar market, consumer surplus will _____, producer surplus will _____, and total surplus will _____.

- A) increase; decrease; decrease
- B) increase; decrease; increase
- C) **decrease ; increase; increase**
- D) decrease; decrease; decrease

10.) Tariffs and import quotas both result in

- A) higher levels of domestic consumption.
- B) the domestic government gaining revenue.
- C) lower levels of domestic production.
- D) **lower levels of imports**

11. Consider the following graph for the market for shirts in Germany, where Germany is importing shirts. **How are domestic consumers and producers impacted in terms of surplus?**



Ans 11.

CS [at local price] = $0.5 \times (100 - 50) \times 150 = \3750

PS [at local price] = $0.5 \times (50 - 20) \times 150 = \2250

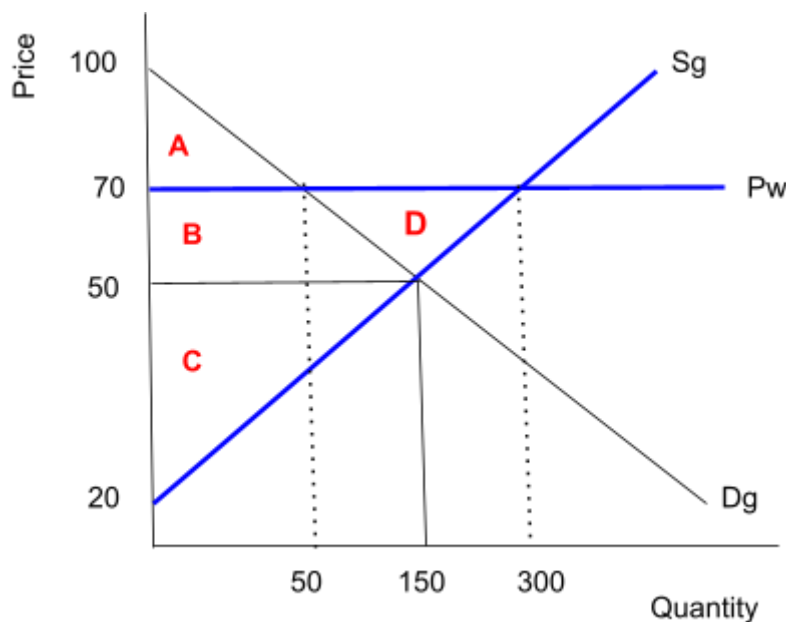
CS [at world price] = $0.5 \times (100 - 30) \times 300 = \10500 [the big triangle]

PS [at world price] = $0.5 \times (30 - 20) \times 50 = \250 [the small triangle]

In terms of surplus, CS increases and PS decreases after Germany imports shirts.

- Increase in the total surplus = area of triangle D = $0.5 \times (300 - 50) \times (50 - 30) = \2500
- ★ Area A is the initial CS
- ★ Area B and C are the initial PS
- ★ Area B is the transfer of PS to CS
- ★ Area D is the increase in total surplus

12. Consider the following graph for the market for cars in Germany, where Germany is exporting cars. How are domestic consumers and producers impacted in terms of surplus?



Answer 12.

Initial CS = $0.5 \times (100 - 50) \times 150 = \3750

Initial PS = $0.5 \times (50 - 20) \times 150 = \2250

CS [after exporting] = $0.5 \times (100 - 70) \times 50 = \750 [the small triangle]

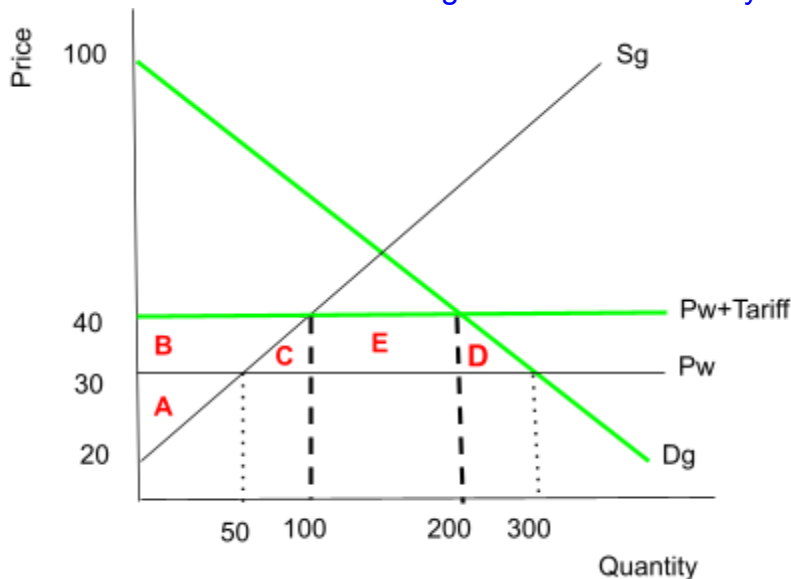
PS [after exporting] = $0.5 \times (70 - 20) \times 300 = \7500 [the big triangle]

In terms of surplus, CS decreases and PS increases after Germany exports cars.

- Increase in the total surplus = area of triangle D = $0.5 \times (300 - 50) \times (70 - 50) = \2500
- ★ Area A and B are the initial CS
- ★ Area C is the initial PS
- ★ Area B is the transfer of CS to PS
- ★ Area D is the increase in total surplus

13. Consider the following graph for the market for shirts in Germany, where Germany is importing shirts. Suppose the German government decides to impose a tariff on imported shirts.

- What will be the consumer and producer surplus after the imposition of tariff?
- Calculate tariff revenue for the German government.
- Calculate deadweight loss for the society.



Answer 13.

- Identify the new QD and QS at the world price+tariff

CS [at world price] = $0.5 \times (100 - 30) \times 300 = \10500

PS [at world price] = $0.5 \times (30 - 20) \times 50 = \250

CS [at world price+tariff] = $0.5 \times (100 - 40) \times 200 = \6000

PS [at world price+tariff] = $0.5 \times (40 - 20) \times 100 = \1000

In terms of surplus, PS increased and CS decreased after the imposition of tariff.

- tariff revenue = tariff amount $\times$ imported quantity

Tariff revenue = rectangular area E = $(40 - 30) \times (200 - 100) = \1000

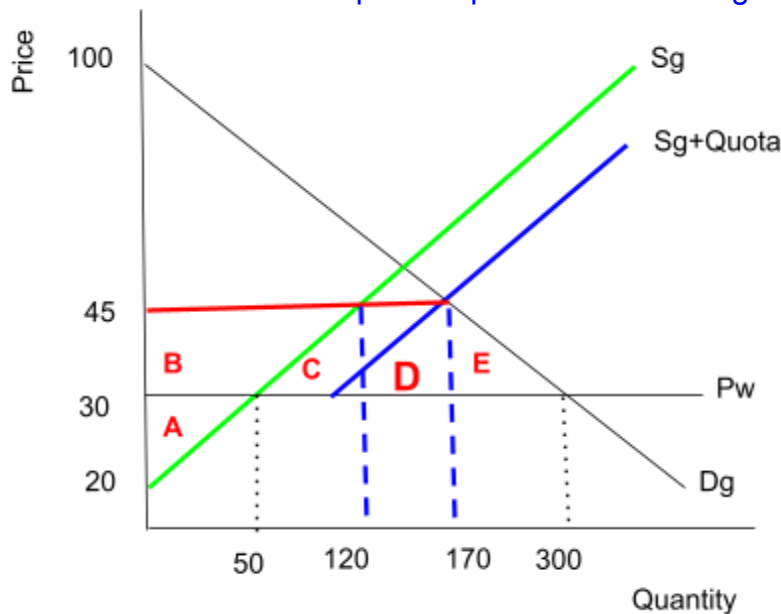
- Deadweight loss from tariff = triangular area C + triangular area D = $\{0.5 \times (40 - 30) \times (100 - 50)\} + \{0.5 \times (40 - 30) \times (300 - 200)\} = 250 + 500 = \750

★ 3 consequences of imposing a tariff-

- The government now has a tax revenue
- Imports are now more expensive so there is a decrease in the quantity of imports
- Domestic production of the imported good increases

14. Consider the following graph for the market for shirts in Germany, where Germany is importing shirts. Suppose the German government decides to impose a quota on imported shirts.

- What will be the consumer and producer surplus after the imposition of tariff?
- Calculate importers' profit and deadweight loss for the society.



Answer 14.

a. Identify the new QD and QS after imposition of import quota

$$CS \text{ [at world price]} = 0.5 \times (100 - 30) \times 300 = \$10500$$

$$PS \text{ [at world price]} = 0.5 \times (30 - 20) \times 50 = \$250$$

$$CS \text{ [import quota]} = 0.5 \times (100 - 45) \times 170 = \$4675$$

$$PS \text{ [import quota]} = 0.5 \times (45 - 20) \times 120 = \$1500$$

In terms of surplus, PS increased and CS decreased after the imposition of tariff.

$$b. \text{ Importers' profit} = \text{rectangular area D} = (45 - 30) \times (170 - 120) = \$750$$

$$\text{Deadweight loss from tariff} = \text{triangular area C} + \text{triangular area E} = \{0.5 \times (45 - 30) \times (120 - 50)\} + \{0.5 \times (45 - 30) \times (300 - 170)\} = 525 + 975 = \$1500$$

★ 3 consequences of imposing an import quota-

- Domestic production of the good increases
- Price of the good increases
- There is a deadweight loss

Note: All these problems have been taken from external sources.

